Physics - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

# Multiple Choice (10 marks)

1. In the atomic nucleus, electrical forces tend to
   (a) cause particles to become neutral
   (b) increase the mass of the nucleus
   (c) cause particles to spin
   (d) generate heat within the nucleus
   (e) create magnetic fields

2. A car battery supplies a currentI of 50 amp to the starter motor. How much charge passes through the starter in (1/2) min?
   (a) 1000 coul
   (b) 1800 coul
   (c) 3000 coul
   (d) 1500 coul
   (e) 2500 coul

3. Electricity and magnetism connect to form
   (a) mass
   (b) low-frequency sound
   (c) nuclear power
   (d) energy
   (e) heat

4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T. The average total energy of the oscillator is
   (a) 2 kT
   (b) (7/2) k T
   (c) 5 kT
   (d) (1/4) k T
   (e) 3 kT

5. An airplane lands on a carrier deck at 150 mi/hr and is brought to a stop uniformly, by an arresting device, in 500 ft. Find the acceleration and the time required to stop.
   (a) -45 ft/$sec^2$ and 4 sec

(b) -48.4 ft/$sec^2$ and 4.55 sec
(c) -40 ft/$sec^2$ and 6 sec
(d) -42 ft/$sec^2$ and 5.5 sec
(e) -52 ft/$sec^2$ and 4.8 sec

# True / False (10 marks)

*Answer True or False*

6. True or False: The total energy needed to remove both electrons from a helium atom is 39.5 eV.

7. True or False: Solar radiation is the primary contributor to Earth's internal energy, primarily through thermal conduction and gravitational pressure.

8. True or False: The mass of an atom comes mostly from its electrons, and its volume is primarily due to its protons.

9. True or False: Protons have a half-life of $10^33$ years, resulting in a decay rate of approximately 4.8 e-05 per day for a 62-kg person.

10. True or False: A car engine with a power output of 33 horsepower generates a force of 300 lbs while traveling at 45 mi/hr.

# Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the motion of a stone thrown horizontally from a bridge with an initial speed of 10 m/s. Analyze the time it takes to reach the water 80 m below, considering gravitational effects and neglecting air resistance.

12. Analyze the motion of passengers on a carnival ride moving in a horizontal circle of radius 5.0 m, completing a revolution in 4.0 s. Calculate their acceleration and discuss the implications of this acceleration in terms of centripetal force and safety.

*End of Paper*